Horizontal-Well Pressure Analysis

Erdal Ozkan,* SPE, and **Rajagopal Raghavan,**** SPE, U. of Tulsa, and **Sadanand D. Joshi,†** SPE, Phillips Petroleum Co.

SPE 16378

Summary. This paper presents an analysis of the pressure-transient behavior of a horizontal well or a drainhole. The performances of horizontal wells and fully penetrating vertical fractures are compared. Dimensionless wellbore pressures are computed for two classic boundary conditions: infinite conductivity and uniform flux. Results are presented as pseudoskin factors and as type curves. In addition to conventional pressure-vs.-time type curves, derivative type curves from pressure/time predictions are presented. The derivative approach we discuss is applicable to a broader range of problems than that considered here.

Computations suggest that horizontal-well productivity is governed by two parameters: (1) the dimensionless well length, L_D $\left[L_D=L/(2h)\sqrt{k_z/k}\,\right]$, and (2) the dimensionless well radius, r_{wD} (the ratio of well radius to well half-length if the formation is assumed to be isotropic). Results indicate that horizontal-well productivity (infinite-conductivity idealization) is almost identical to that of a fully penetrating vertical fracture of infinite conductivity if $L_D>4.0$. This result shows that horizontal wells may perform better than vertically fractured wells if nonideal aspects associated with vertically fractured wells, such as limited conductivity or height, are taken into account.

Results have been obtained for the range $0.1 \le L_D \le 100$ and for the range $10^{-4} \le r_{wD} \le 10^{-2}$; these ranges should encompass expected values of horizontal-well length, reservoir height, and the ratio of vertical and horizontal permeability. Pressure responses and pseudoskin factors are calculated for four different well locations within the productive interval. The pseudoskin factors are insensitive to well location.

Introduction

A review of production performance over the past few years conclusively establishes the advantages of horizontal wells and drainholes. Drainholes are normally drilled from existing vertical wells and extend 100 to 500 ft [30 to 152 m] in either direction. Horizontal wells involve the drilling of new wells and are usually 1,000 to 2,000 ft [305 to 610 m] long. Both completions, like fractured wells, are intended to provide a larger surface area for fluid withdrawal and thus improve productivity. These completions have been found to be effective in (1) some naturally fractured reservoirs, (2) reservoirs where gas- and/or water-coning problems preclude the efficient operation of vertical wells, (3) thin reservoirs, and (4) reservoirs with high vertical permeability. Disadvantages of these completions at the present time include high drilling costs and the inability to produce contiguous zones that are separated by impermeable layers through a single wellbore.

For modeling well behavior, horizontal wells and drainholes can be treated similarly. Thus, unless specifically stated, the term horizontal well will include both completion types.

To be an economical completion method, at least for primary recovery, horizontal-well productivity must be comparable with the productivity of a vertically fractured well. We agree with others[1,2] that it is not sufficient to compare horizontal-well productivity with unstimulated vertical wells. Thus, in this work, responses for vertically fractured wells are used as a basis for comparison. (A horizontal well may be viewed as a fracture with limited height; however, this does not imply that horizontal-well productivity will never be comparable with that of a vertically fractured well.)

The transient behavior of horizontal wells and the characteristics of response curves have been considered in Refs. 3 through 5. The central new results of this work concern the development of analytical expressions and correlations for the pseudoskin factor and of new analysis methods to determine formation properties and well characteristics. In particular, we discuss new applications of the derivative approach—the pressure normalized by its derivative.[6] This procedure was first presented in the groundwater-hydrology literature by Chow.[6] The procedure we suggest has a much broader range of applicability than is shown.

Problem Formulation

The mathematical model considered is identical to that examined in Refs. 3 and 5. We consider the flow of a slightly compressible fluid to a horizontal well of length L in a reservoir of height h. The well is assumed to be parallel to the top and bottom boundaries, and gravity effects are considered to be negligible. We consider an anisotropic system with permeabilities k_x, k_y, and k_z. The well is assumed to be located at any location, z_w, within the vertical interval and is considered to be a line-source (Fig. 1). Two boundary conditions on the well surface are considered: infinite conductivity and uniform flux. The consequences of using these boundary conditions are discussed later.

The reservoir pressure distribution is given by the following equation[3]:

$$p_D(x_D,y_D,z_D,z_{wD},L_D,t_D)$$
$$=\frac{\sqrt{\pi}}{4}\sqrt{\frac{k}{k_y}}\int_0^{t_D}\left[\operatorname{erf}\frac{\left(\sqrt{k/k_x}+x_D\right)}{2\sqrt{\tau}}+\operatorname{erf}\frac{\left(\sqrt{k/k_x}-x_D\right)}{2\sqrt{\tau}}\right]$$
$$\times[\exp(-y_D^2/4\tau)]$$
$$\times\left[1+2\sum_{n=1}^{\infty}\exp(-n^2\pi^2L_D^2\tau)\cos n\pi z_D\cos n\pi z_{wD}\right]\frac{d\tau}{\sqrt{\tau}}. \quad (1)$$

Here, p_D = dimensionless pressure at any point in the reservoir and t_D = dimensionless time based on $L/2$. p_D and t_D are defined by

$$p_D(x_D,y_D,z_D,z_{wD},L_D,t_D)$$
$$=kh/141.2qB\mu[p_i-p(x,y,z,z_w,L,t)] \quad (2)$$

and $t_D=0.001055kt/\phi c_t\mu L^2$. ......(3)

In Eqs. 1 through 3, k is an arbitrary constant and may be chosen to be the permeability of an equivalent isotropic system $\left(k=\sqrt[3]{k_xk_yk_z}\right)$ or to be the equivalent horizontal permeability $\left(k=\sqrt{k_xk_y}\right)$. One could even assume that $k=k_x$ because its definition is dictated by the system under study. In this paper, unless otherwise mentioned, we define k as the equivalent horizontal permeability $\left(k=\sqrt{k_xk_y}\right)$.

Dimensionless distances x_D and y_D are based on $L/2$, and z_D is based on h. The following relations define x_D, y_D, and z_D, respectively:

$$x_D=(2x/L)\sqrt{k/k_x}, \quad (4)$$

$$y_D=(2y/L)\sqrt{k/k_y}, \quad (5)$$

and $z_D=z/h$. ......(6)

The center of the well is assumed to be at $(0,0,z_w)$, and L_D is given by

$$L_D=(L/2h)\sqrt{k_z/k}. \quad (7)$$

Note that L_D incorporates the anisotropy of the formation. If the formation is isotropic and $L_D=0.5$, then the surface area available for fluid production is identical to that of a fully penetrating vertical well.

*Now at Istanbul Technical U.
**Now at Texas A&M U.
†Now at Joshi Production Technologies Inc.

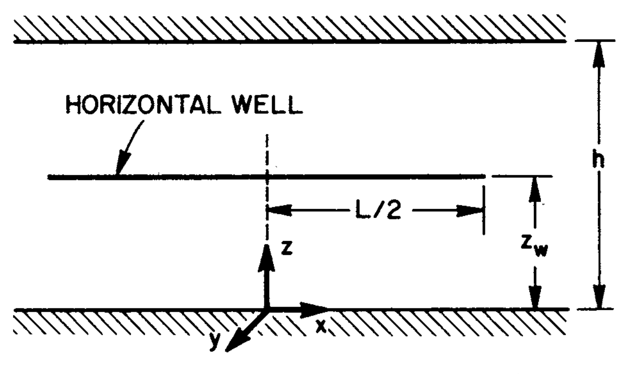

Fig. 1—Schematic of system examined in this study.

The dimensionless variables given in Ref. 5 are different from those given above. Eq. 9 of Ref. 5 can be obtained from Eq. A-6 of the same reference if $k_x = k_y$. Thus, their work assumes that $k_x = k_y = k$.

Asymptotic Forms of Eq. 1. Short- and long-time approximations of Eq. 1 can be derived with the methods given in Refs. 7 and 8. The short-time approximation of Eq. 1 is given by[5]

$$p_D(x_D,y_D,z_D,z_{wD},L_D,t_D)$$

$$= -\frac{\beta}{8L_D}\mathrm{Ei}\left[-\frac{(z_D-z_{wD})^2/L_D^2+y_D^2}{4t_D}\right], \quad (8)$$

where $\beta=2$ for $|x_D|<\sqrt{k/k_x}$, $\beta=1$ for $|x_D|=\sqrt{k/k_x}$, and $\beta=0$ for $|x_D|>\sqrt{k/k_x}$.

The duration for which Eq. 8 is valid is a function of x_D, z_D, z_{wD}, and L_D. The duration of this initial radial-flow period is given by[5]

$$t_D \le \min\begin{cases}\delta_D^2/20, \\ (z_D+z_{wD})^2/(20L_D^2), \\ (z_D+z_{wD}-2)^2/(20L_D^2),\end{cases} \quad (9)$$

where $\delta_D = \sqrt{k/k_x} - |x_D|$ if $|x_D|<\sqrt{k/k_x}$, and $\delta_D = 2\sqrt{k/k_x}$ if $|x_D| = \sqrt{k/k_x}$.

These expressions are *approximate* and the actual times for which Eq. 8 is valid can be determined only with computations. If we assume $k_x \ne k_y \ne k_z \ne k_x$, then the counterpart of Eq. 5 in dimensional form for $|x|<L/2$ is given by

$$\sqrt{k_y k_z}\,L[p_i - p(|x|<L/2,y,z,z_w,t)]/141.2qB\mu$$

$$= -\frac{1}{2}\mathrm{Ei}\left[-\frac{y^2k_z+(z-z_w)^2k_y}{4(0.0002637\eta_y\eta_z t)}\right], \quad (10)$$

where $\eta_j = k_j/(\phi c_t \mu)$, where $j=z$ or y. Eq. 10 suggests that the early-time response will be identical to that of a vertical well in a formation with $h=L$; i.e., the well behaves as if it is located in an areally infinite reservoir of height L (infinite-acting radial flow). If this flow period can be identified on a well test, then it is possible to determine $\sqrt{k_y k_z}$ from the slope of the semilog straight line. An examination of Eqs. 8 through 10 indicates that this initial radial-flow period ends when the effect of the closest physical boundary (the top or the bottom reservoir boundary) is felt or when flow across the well tips ($x=\pm L/2$) affects the pressure response. Detailed information on the duration of this flow period follows.

At long times, the pressure distribution is given by

$$p_D(x_D,y_D,z_D,z_{wD},L_D,t_D) = ½(\ln t_D + 2.80907)$$

$$+\sigma(x_D,y_D)+F(x_D,y_D,z_D,z_{wD},L_D), \quad (11)$$

where $\sigma(x_D,y_D)$ and $F(x_D,y_D,z_D,z_{wD},L_D)$ are given, respectively, by

$$\sigma(x_D,y_D) = 0.25\Big\{\big(x_D-\sqrt{k/k_x}\big)\ln\big[\big(x_D-\sqrt{k/k_x}\big)^2+y_D^2\big]$$

$$-\big(x_D+\sqrt{k/k_x}\big)\ln\big[\big(x_D+\sqrt{k/k_x}\big)^2+y_D^2\big]$$

$$-2y_D\arctan\big[2y_D/\big(x_D^2+y_D^2-\sqrt{k/k_x}\big)\big]\Big\} \quad (12)$$

and $F(x_D,y_D,z_D,z_{wD},L_D)$

$$= \sum_{n=1}^{\infty}\cos n\pi z_D \cos n\pi z_{wD}\int_{-1}^{+1}K_0(\tilde{r}_D L_D n\pi)\mathrm{d}\alpha, \quad (13)$$

where $\tilde{r}_D^2 = \big(x_D - \alpha\sqrt{k/k_x}\big)^2 + y_D^2$. (14)

Eq. 11 is derived in Appendix A of Ref. 9 and in Ref. 10. If $F=0$, then Eq. 11 yields the long-time pressure distribution caused by a fully penetrating uniform-flux fracture (see Eq. 14 of Ref. 8). As shown in Appendix A of Ref. 9, Eq. 11 can describe the pressure response when t_D is given by[10]

$$t_D \ge \max\begin{cases}100/(\pi L_D)^2 \\ 25\big[\big(x_D-\sqrt{k/k_x}\big)^2+y_D^2\big] \\ 25\big[\big(x_D+\sqrt{k/k_x}\big)^2+y_D^2\big]\end{cases} \quad (15)$$

Eq. 11 indicates that at long times a semilog plot of p_D vs. t_D will be a straight line with a slope equal to 1.151—the pseudoradial-flow period. If data during this flow period are available, then formation permeability, k or $\sqrt{k_x k_y}$—in case we consider anisotropy in the x, y plane—can be determined. Formation permeability, k, can also be obtained by type-curve matching. If k_z is known, and the initial radial-flow period can be analyzed to obtain $\sqrt{k_y k_z}$, then the possibility to determine k_x, k_y, and k_z exists. Eq. 11 also permits us to obtain an analytical expression for the pseudoskin factor (see below).

It must be emphasized that the restrictions given by Eq. 15 are approximate; different arrangements of Eq. 1 and the use of different approximations during the derivations will lead to the expression given by Eq. 11, but Eq. 15 would be different. Our calculations indicate that the time restrictions given by Eq. 15 provide the highest order of accuracy, especially in computing the pseudoskin factor.

Computation of Well Responses. Without loss of generality, we assume that the formation is isotropic, and we model the horizontal well as a line source. First, we note the similarity between the short-time approximation for the horizontal-well solution given by Eq. 10 and the line-source approximation for a vertical well in an infinite reservoir. Then, by analogy to the vertical-well solution, we may conclude that the responses predicted by Eq. 8 would approximate pressures of a horizontal well of finite radius when

$$t_D \ge 25[(z_D-z_{wD})^2/L_D^2+y_D^2]. \quad (16)$$

This requirement will be satisfied for most practical applications of well testing. We also note (from Eq. 8) that if we define a circle with the origin at $y_D = z_D - z_{wD} = 0$ and

$$r_D = \sqrt{(z_D-z_{wD})^2/L_D^2+y_D^2} = 2r_w/L, \quad (17)$$

then the pressure drop calculated at any point on the circumference of this circle will be the same at early times. This result suggests that we define r_{wD} from Eq. 17 and calculate the well responses at $r_D = r_{wD}$. It is not obvious from the long-time approximation of the horizontal-well solution (Eq. 11) that this definition of the wellbore radius would be correct for all times. However, one can readily decide that when the line-source assumption for the finite-radius horizontal well becomes acceptable, then the error introduced in the definition of the wellbore-pressure measurement point would not have an influence on the accuracy of the results. Our computations for dimensionless pressure defined by Eq. 1 indicate that the dimensionless pressure never changes more than 0.005 along the circumference of the wellbore defined by Eq. 17.

In Refs. 3 and 5, wellbore pressures are computed at the point $z_D=z_{wD}$ and $y_D=r_{wD}$. Eq. 17 and the discussion following it, however, indicate that the point $y_D=0$ and $(z_D-z_{wD})/L_D=r_{wD}$ also could be chosen to compute wellbore responses. As already mentioned, numerical computations prove that the wellbore pressures computed at $z_D=z_{wD}$ and $y_D=r_{wD}$ or $(z_D-z_{wD})/L_D=r_{wD}$ and $y_D=0$ are essentially the same. Here, we use both definitions to compute the wellbore responses. Mainly for comparison with the published data,[3,5] we choose the point $z_D=z_{wD}$ and $y_D=r_{wD}$ to compute the well responses when we present our results. As will be shown later, the use of the point $z_D=z_{wD}+r_{wD}L_D$ and $y_D=0$ is advantageous for the evaluation of the analytical expressions for the long-time approximation and the pseudoskin factor. For convenience, we define the dimensionless wellbore radius by $r_{wzD}=r_{wD}L_D$. Note that, in terms of real variables, r_{wD} and r_{wzD} are defined respectively by

$$r_{wD}=(2r_w/L)\sqrt{k/k_y} \quad \text{(18)}$$

and $r_{wzD}=r_{wD}L_D=(r_w/h)\sqrt{k_z/k_y}$. .... (19)

Therefore, a simple conversion exists between the results computed at r_{wD} and r_{wzD}.

To simulate an infinite-conductivity wellbore, x_D was chosen to be 0.732. Justification for this value is based on the work of Gringarten *et al.*,[8] who showed that the uniform-flux solution can be used to compute the response of a well intercepted by an infinite-conductivity vertical fracture if $x_D=0.732$. This choice can also be justified on the basis of Eq. 11. As already mentioned, a horizontal well may be viewed as a vertical fracture where the height of the fracture tends to zero in the limit. If the solutions for an infinite-conductivity wellbore are to be in agreement with the infinite-conductivity, vertical-fracture solutions during the pseudoradial-flow period, then from Eq. 11, we require that $x_D=0.732$. (Ref. 11 presents equivalent pressure points for horizontal wells. Intuitively, we expect the equivalent pressure point to be a function of anisotropy and well location with respect to the boundaries; the results in Ref. 11 confirm this view.) The infinite-conductivity solution can be used to analyze well responses for either drainhole or horizontal-well completion.

To compute well responses for the uniform-flux case, we assumed that $x_D=0$. This solution should be applicable to drainholes that extend equal distances on either side of the vertical well. The uniform-flux solution does not appear to be appropriate for analyzing horizontal-well responses. If we assume that the uniform-flux solution is valid for the horizontal-well case, then we should assume that the pressure at $|x|=L/2$ would represent the wellbore pressure. (Note that the uniform-flux solution, Eq. 1, predicts that pressure will vary along the length of the well.) The pressure is the lowest at the well center ($x_D=0$). Although the method of sources and sinks[7] used to develop Eq. 1 provides no information regarding the ultimate extraction of fluid once fluid enters the vertical fracture or horizontal well, the assumption that pressure is measured at the tip should imply that fluid from the reservoir is extracted at this point. This assumption will result in the unrealistic implication that fluid can move from a region of lower pressure ($x_D=0$) to higher pressure ($|x|=L/2$).

The fact that the wellbore pressure is computed at a finite radius, r_w, has ramifications that deserve consideration. The vertical-fracture solutions given in the literature ignore the existence of the wellbore. It is possible to compute the response for a vertically fractured well at $x_D=0$, $y_D=0$ and specify the pressure at this point to be the wellbore pressure. (Mathematically this implies that it is possible to compute pressures within the source and that these solutions are bounded for all time values.) Because the horizontal-well case is assumed to be a line source, however, it is not possible to compute pressure drops on the source. Pressure drops have to be computed at some finite radius, r. Thus, consideration must be given to two factors in the analysis of computations. First, the horizontal-well solutions are a function of r_{wD}. (This is also true for fractured wells and usually is ignored.) Second, because the vertical-fracture solutions and the horizontal-well solutions are not computed at the same point, it can be shown that in some cases the productivity of a horizontal well will exceed that of a vertically fractured well, all other conditions being identical. This unrealistic result is obtained mainly because of the locations where the pressure drops are computed and should not be construed to imply that horizontal-well productivity can be greater than that of a vertically fractured well. Other considerations (reservoir heterogeneity, fluid contacts, etc.) may result in horizontal-well productivity being higher than that of a vertically fractured well.[12] Note that the latter consideration cannot be avoided even if we assume that the horizontal well has a finite radius.

Pseudoskin Factor—Analytical Expression. Pseudoskin factors, S, considered here are based on the appropriate vertically-fractured-well solution. In light of the above discussion, we chose the point $y_D=0$, $z_D=z_{wD}+r_{wzD}$ to compute the pseudoskin factor. From the expression for $F\left(|x_D|\le\sqrt{k/k_x},0,z_{wD},r_{wzD},L_D\right)$ derived in Appendix A,[9] we have

$$S\equiv F\left(|x_D|\le\sqrt{k/k_x},0,z_{wD},r_{wzD},L_D\right)$$
$$=-\frac{1}{2L_D\sqrt{k/k_x}}\ln\left\{4\sin\left[\frac{\pi}{2}(2z_{wD}+r_{wzD})\right]\sin\left(\frac{\pi}{2}r_{wzD}\right)\right\}$$
$$-\varphi(x_D,0,z_{wD},r_{wzD},L_D), \quad \text{(20)}$$

where $\varphi(x_D,0,z_{wD},r_{wzD},L_D)$

$$=\frac{1}{\pi L_D\sqrt{k/k_x}}\sum_{n=1}^{\infty}\frac{\cos n\pi(z_{wD}+r_{wzD})\cos n\pi z_{wD}}{n}$$
$$\times\left\{Ki_1\left[n\pi L_D\left(\sqrt{k/k_x}+x_D\right)\right]+Ki_1\left[n\pi L_D\left(\sqrt{k/k_x}-x_D\right)\right]\right\}. \quad \text{(21)}$$

Here, $Ki_1(x)=\int_x^{\infty}K_0(u)\mathrm{d}u$. .... (22)

To the best of our knowledge, the expression for the horizontal-well pseudoskin factor given by Eq. 20 has not been reported in the literature. As will be discussed later, our computations indicate that for $L_D\ge 4$, F becomes negligible and the differences between the late-time responses of vertically fractured and horizontal wells vanish for practical purposes. In physical terms, the logarithmic term on the right side of Eq. 20 represents the contribution of the early-time radial-flow period in the vertical plane (see Eq. 8) to the pseudoskin factor. Similarly, the terms denoted by φ (Eq. 21) represent the contribution of the 3D flow period between the early-time radial flow and late-time pseudoradial-flow periods. As mentioned earlier, the early-time flow period in the vertical plane (y, z plane) ends either when the top or bottom boundary influences the well response or when flow across the well tips affects the well response. If this flow period ends as a result of the influence of the boundaries, then the resistance to flow in the vertical direction becomes negligible compared with that in the horizontal direction. If flow across the well tips distorts the early-time flow period, then a 3D flow period exists and the function φ is not negligible. The conditions under which the 3D flow period contributes significantly to the pseudoskin factor can be obtained by examining Eq. 21. Note that if φ is negligible, then for all practical purposes 2D flow regimes—in the vertical (y, z) and in the horizontal (x, y) planes at early and late times, respectively—dominate the well response.

If we note that $Ki_1(x)$ can be neglected when $x\ge 3.6$, then the function φ becomes negligible compared with the first term on the right side of Eq. 20, within 1%, when

$$L_D\ge 3.6/\left[\pi\left(\sqrt{k/k_x}-|x_D|\right)\right]. \quad \text{(23)}$$

The long-horizontal-well assumption has been extensively used in the literature to obtain approximate solutions for the pressure distribution and to derive expressions for the pseudoskin factor. The fact that the 3D flow period (the function φ) is negligible when L_D satisfies the condition given by Eq. 23 is the basis for the steady-state solution obtained by Joshi.[13] An expression for the pseudoskin factor (which is strictly applicable for $z_w=h/2$) also is given in Ref. 3 (Eq. A-5). Because this equation is not derived, it is not possible to comment on the relation given in Ref. 3; however, we believe that assumptions similar to those given by Joshi were used

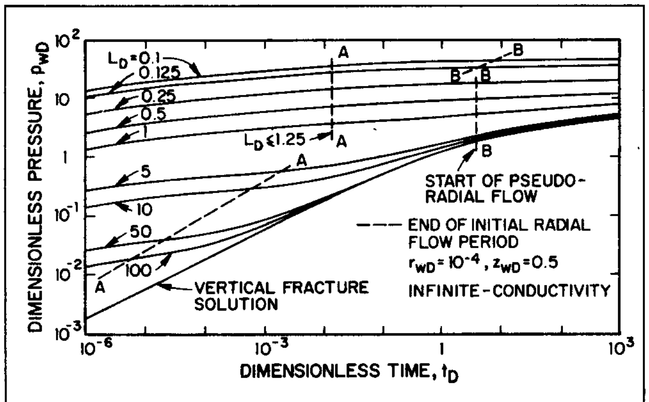

Fig. 2—Pressure response for horizontal well—infinite conductivity.

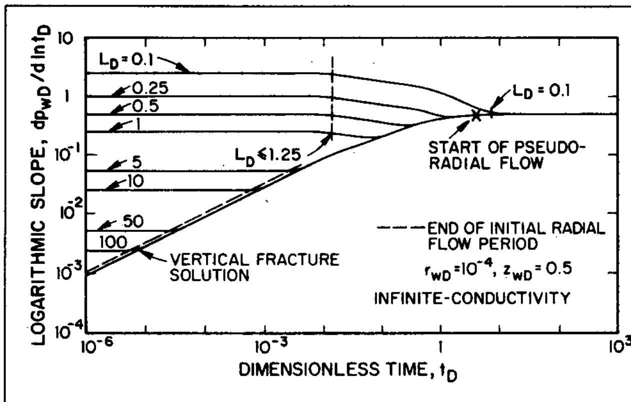

Fig. 3—Derivative response for horizontal well—infinite conductivity.

to derive Eq. A-5 of Ref. 3. On the basis of this discussion, it can be shown that when the condition given by Eq. 23 holds, we can approximate the horizontal-well pseudoskin factor by

$$S=F\left[|x_D|\le\sqrt{k/k_x},0,z_{wD},r_{wzD},L_D\ge 1.15/\left(\sqrt{k/k_x}-|x_D|\right)\right]$$

$$\approx-\frac{1}{2L_D\sqrt{k/k_x}}\ln\left\{4\sin\left[\frac{\pi}{2}(2z_{wD}+r_{wzD})\right]\sin\left(\frac{\pi}{2}r_{wzD}\right)\right\}. \quad (24)$$

There is field evidence that most horizontal-well lengths would satisfy the condition stated by Eq. 23 and, therefore, the use of the approximate relation given by Eq. 24 would be satisfactory for the computation of the pseudoskin factor.

As discussed above, if $L_D\ge 4$, then F is negligible and therefore the long-time responses of vertically fractured wells and horizontal wells are practically the same. This result suggests that the equivalent pressure point $x_D=0.732$ for a vertically fractured well also can be used for long horizontal wells ($L_D\ge 4$). We have shown also that the function φ becomes negligible if $L_D\ge 1.15/\left(\sqrt{k/k_x}-|x_D|\right)$ and the long-time response of a horizontal well is given by

$$p_{wD}\left[L_D\ge 1.15/\left(\sqrt{k/k_x}-|x_D|\right)\right]=p_{fD}$$

$$-\frac{1}{2L_D\sqrt{k/k_x}}\ln\left\{4\sin\left[\frac{\pi}{2}(2z_{wD}+r_{wzD})\right]\sin\left(\frac{\pi}{2}r_{wzD}\right)\right\}. \quad (25)$$

Note that the second term on the right side of Eq. 25 (the horizontal-well pseudoskin factor) is independent of flow in the x direction. This suggests that the point $x_D=0.732$, obtained by Ref. 8 for vertically fractured wells, can be used as the equivalent pressure point for horizontal wells if $L_D\ge 1.15/\left(\sqrt{k/k_x}-|x_D|\right)$.

Note that it is possible to compute the equivalent pressure point for horizontal wells as was done for vertically fractured wells.[8] In fact, the equivalent pressure point for horizontal wells is presented in Ref. 11 by computing the stabilized flux distribution as outlined in Ref. 8. Ref. 11 presents the equivalent pressure points for the range of horizontal-well lengths $2.5\le L_D\le 20$. The values of the equivalent pressure point given in Ref. 11 appear to be in conflict with our observation that $x_D=0.732$ for $L_D\ge 1.15/\left(\sqrt{k/k_x}-|x_D|\right)$. Their computations also do not appear to be internally consistent.

Pressure Derivative Analysis. The advent of sensitive pressure gauges makes it possible to use the pressure derivative as a practical analysis method. We present an alternative approach of analyzing well-test data with the derivative method that involves plotting $\Delta p/[2\partial(\Delta p)/\partial\ln t]$ with respect to time. This procedure was first presented in the groundwater literature in 1952.[6] This method has a much wider range of applicability and should improve our ability to analyze data, particularly with regard to heterogeneous reservoirs. In this paper, however, we restrict our attention to the problem under consideration and present applications for three cases: the exponential-integral, horizontal-well, and vertically-fractured-well solutions. Detailed derivations are given in Ref. 10 and are summarized in Appendix B of Ref. 9. Here, we briefly outline details.

For simplicity, let us consider the exponential-integral solution:

$$p_D(r_D,t_D)=-\tfrac{1}{2}\mathrm{Ei}[-r_D^2/(4t_D)]. \quad (26)$$

Noting that $\partial p_D/\partial\ln t_D=0.5\exp[-r_D^2/(4t_D)]$, we have

$$\frac{p_D(r_D,t_D)}{2(\partial p_D/\partial\ln t_D)}=-0.5\exp\left(\frac{r_D^2}{4t_D}\right)\mathrm{Ei}\left(-\frac{r_D^2}{4t_D}\right). \quad (27)$$

For large times, $[r_D^2/(4t_D)<0.01]$, $\exp[r_D^2/(4t_D)]\rightarrow 1$ and we have

$$\frac{p_D(r_D,t_D)}{2(\partial p_D/\partial\ln t_D)}\approx p_D(r_D,t_D)=0.5\left[\ln\frac{t_D}{r_D^2}+0.8091\right]. \quad (28)$$

Eqs. 26 through 28 teach us the following. First, if the semilog straight line exists, then a plot of $\Delta p/(2\partial\Delta p/\partial\ln t)$ vs. log t should be a straight line with slope equal to 1.151. Thus, this procedure permits us to identify the semilog straight line in a straightforward manner, should it exist, and can be used for any wellbore condition. Second, it is possible to prepare a type curve with Eq. 27 and use it to match data in the conventional way. In this case, we would plot $\Delta p/(2\partial\Delta p/\partial\ln t)$ vs. t. If the semilog straight line does not exist, then it is possible to determine hydraulic diffusivity, η, by aligning the vertical axes of the type and field curves. (Note that the vertical axes for the type and field curves are dimensionless.) Once η is determined, pressure data can be used to match the well response to obtain kh (the time axes are now fixed) and the $\phi c_t h$ product can be determined from the estimates of kh and η. Although we have considered the exponential-integral solution only, this method is applicable to other situations. Ref. 14 presents field applications of this procedure.

Results

Infinite-Conductivity Solutions. Fig. 2 shows typical pressure responses of infinite-conductivity horizontal wells. The variable of interest is the dimensionless horizontal-well length, L_D ($0.1\le L_D\le 100$). The solutions shown here are for a well located at the reservoir mid-height ($z_{wD}=0.5$) and $r_{wD}=10^{-4}$. It is also assumed that $k=k_x=k_y$. The bottom curve in Fig. 2 is the response of a fully penetrating vertically fractured well. Fig. 2 shows that the $L_D\ge 50$ solutions are indistinguishable from the vertically-fractured-well solution (on log-log coordinates) for $t_D\ge 2.3\times 10^{-1}$ (relative difference $\le 5\%$); i.e., at long times, the pressure responses of long horizontal wells are almost identical to the responses for fully penetrating vertically fractured wells. Our computations indicate that the difference between the horizontal-well and vertically-fractured-well responses at late times is $<10\%$ if $L_D\ge 4$. The dashed lines on the left side, lines marked AA in Fig. 2, denote

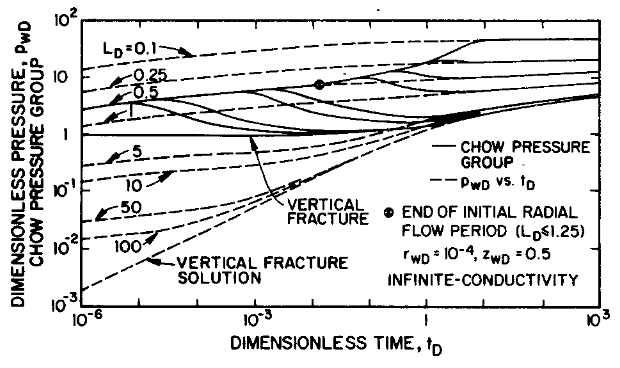

Fig. 4—Composite type curve, pressure response, and derivative response—infinite conductivity.

TABLE 1—THE INFLUENCE OF WELL RADIUS ON PRESSURE RESPONSE OF AN INFINITE-CONDUCTIVITY HORIZONTAL WELL ($L_D = 10$, $z_{wD} = 0.5$)

Dimensionless Time, t_D	Dimensionless Wellbore Pressure, p_{wD} — Dimensionless Well Radius, r_{wD}		
	5×10^{-5}	10^{-4}	5×10^{-4}
10^{-6}	0.1700	0.1354	0.05642
10^{-5}	0.2276	0.1929	0.1126
10^{-4}	0.2851	0.2505	0.1700
10^{-3}	0.3439	0.3093	0.2288
10^{-2}	0.4646	0.4300	0.3495
10^{-1}	0.7827	0.7480	0.6675
1	1.491	1.457	1.376
10	2.556	2.521	2.440
10^2	3.697	3.662	3.582
10^3	4.847	4.813	4.732
10^4	5.999	5.964	5.884

the end of the initial radial-flow period (Eq. 8). During this period, the well behaves as if it were a vertical well completely penetrating a formation of thickness L. The end of this flow period was determined by computing slopes, and the lines marked AA reflect times at which slopes differ from 1.151 by 5%. The end of this flow period is determined by two factors. First, movement of fluid across the well tips ($|x_D|=1$) can distort the isopotential lines that are concentric with the well axes. In this case, the end of this flow period will be independent of L_D. Intuitively, we would expect this to be the case for small values of L_D (h large and/or k_z small). This period can end also if upper and/or lower boundaries influence the well response. Under such circumstances, we would expect L_D to have an influence on the end of this period and for the boundaries to influence the well behavior early if L_D is large (h small and/or k_z large). These results are consistent with the time ranges given on the right of Eq. 9.

The lines marked BB denote the beginning of the pseudoradial-flow period. Data beyond this period can be used to obtain formation permeability and skin factor by conventional semilog-analysis techniques. If $L_D \geq 0.25$, the beginning of this flow period is independent of L_D and the upper and lower reservoir boundaries do not influence the beginning of this flow period. The basis for this result can be understood if we examine the derivative of p_{wD}. Fig. 3 is a plot of $dp_{wD}/d \ln t_D$ vs. t_D. Note that the derivative plot has a value equal to $0.25/L_D$ at early times (Eq. 8) and 0.5 at late times (Eq. 11). The letter "x" ($t_D=4.13$) represents the time for the start of pseudoradial flow for horizontal wells for $L_D \geq 0.25$. It also represents the time for the start of pseudoradial flow for a vertically fractured well. This result implies that for $L_D \geq 0.25$, h and k_z/k do not affect the beginning of this flow period; thus, L governs the start of the pseudoradial-flow period and, not surprisingly, this time is identical to the start of the pseudoradial-flow period for a vertically fractured well. If $L_D < 0.25$, then the upper and lower boundaries also control the time for the beginning of pseudoradial flow and thus L_D (vertical permeability and reservoir thickness) has an influence on the beginning of the pseudoradial-flow period. These observations may also be interpreted as follows. If the initial radial-flow period ends because of the influence of the top or bottom boundary (large L_D), then our results show, not unexpectedly, that the start of pseudoradial flow is independent of L_D and is governed by the value of L (with t_D). The reverse argument is also true.

Based on an examination of the responses shown in Figs. 2 and 3, the following conclusions may be derived. First, if L_D is large, then it may not be possible to distinguish the horizontal-well solutions from the vertically-fractured-well solutions. The results shown in Fig. 2 and other computations suggest that if $L_D \geq 50$, the two solutions may be indistinguishable. Second, if L_D is small, then the characteristic shape of the well response will be indistinguishable from that of an unfractured vertical well. The well responses display features similar to that of an unfractured vertical well. If L_D is small enough, type-curve matching of pressure data may be virtually impossible. Also, we may not be able to distinguish the specific radial-flow period (initial or pseudoradial) from the derivative plot if all measured data are taken during one of these flow periods. Fig. 3, however, suggests that the derivative plot would be beneficial if data during the transitional period are available. In the following section, we suggest a procedure to identify the appropriate semilog straight line and to improve our ability to type-curve match pressure data.

The fact that the horizontal-well solutions are, for all practical purposes, identical to the fully penetrating vertical-fracture solution at late times for large L_D values deserves comment. We noted earlier that a horizontal well may be viewed as a vertical fracture of zero height in the limit. Thus, it may be surprising that the two solutions are identical for certain values of L_D. Physically, this result implies that the additional pressure drop as a result of flow in the vertical direction is negligibly small. The magnitude of the additional pressure drop reflecting convergence to the wellbore depends on the manner in which the fluid converges toward the wellbore. At large distances from the well center, velocities are small, and if convergence takes place where velocities are small, then the additional pressure drop that reflects convergence is also small. In the horizontal-well case, fluid begins to converge at large distances from the well and, thus, the additional pressure drop is small. In the case of vertical wells, fluid converges toward the wellbore where velocities are high, and if all conditions are identical (r_w, h, k_z, and k), the corresponding additional pressure drop would be much higher. For this reason, a horizontal well may reduce water- or gas-coning problems (see Refs. 13 and 15).

Although not directly pertinent to this work, the observations in the preceding paragraphs and the results in Figs. 2 and 3 have a bearing on the performance of vertically fractured wells. These results suggest that the long-term productivity of a vertically fractured well is governed principally by the length (and conductivity) of the vertical fracture rather than by the height of the fracture. The results given here imply that if lateral penetration is significant, penetration in the vertical plane is not germane to long-term productivity (see also Ref. 16).

Derivative Responses—Chow Pressure Group. Fig. 4 is a composite of the pressure responses in terms of the Chow pressure group (unbroken lines) and the pressure responses (dashed lines) taken from Ref. 10. At early times, the pressures normalized by their derivative solutions are identical; the influence of L_D is negligible. This result follows directly from Eq. B-17 in Ref. 9. As time increases, the responses in terms of the Chow pressure group diverge and ultimately merge with the appropriate p_{wD} curve (see Appendix B in Ref. 9). Note that the ordinate of the plot in terms of the Chow pressure group does not involve formation or well parameters. This result implies that it is possible to conduct type-curve matching by Chow's method by aligning the vertical scales of the field and type curves (Fig. 4). The circle circumscribing the letter "x" on Fig. 4 reflects the end of the initial radial-flow period for $L_D \leq 1$. From Fig. 4, we note that responses in terms of the Chow pressure group are identical to the pressure responses during the pseudoradial-flow period (Eq. B-25 in Ref. 9). Thus, by type-curve matching data with Chow's method, appropriate semilog straight lines, should they exist, can be identified.

TABLE 2—THE INFLUENCE OF WELL RADIUS ON PRESSURE RESPONSE OF AN INFINITE-CONDUCTIVITY HORIZONTAL WELL ($L_D=1$, $z_{wD}=0.5$)

	Dimensionless Wellbore Pressure, p_{wD}		
Dimensionless	Dimensionless Well Radius, r_{wD}		
Time, t_D	5×10^{-5}	10^{-4}	5×10^{-4}
10^{-6}	1.700	1.354	0.5642
10^{-5}	2.276	1.929	1.126
10^{-4}	2.851	2.505	1.700
10^{-3}	3.427	3.080	2.276
10^{-2}	4.000	3.654	2.849
10^{-1}	4.501	4.155	3.350
1	5.212	4.865	4.060
10	6.276	5.929	5.125
10^2	7.418	7.071	6.266
10^3	8.568	8.221	7.417
10^4	9.719	9.372	8.568

TABLE 3—INFLUENCE OF WELL LOCATION ON PRESSURE RESPONSE OF AN INFINITE-CONDUCTIVITY HORIZONTAL WELL ($L_D=25$, $r_{wD}=10^{-4}$)

	Dimensionless Wellbore Pressure, p_{wD}			
Dimensionless	Well Location, z_{wD}			
Time, t_D	0.5	0.25	0.125	0.0625
2×10^{-6}	0.06109	0.06109	0.06109	0.06119
1×10^{-5}	0.07717	0.07717	0.07742	0.08149
10^{-4}	0.1033	0.1024	0.1106	0.1228
10^{-3}	0.1391	0.1460	0.1583	0.1718
10^{-2}	0.2595	0.2665	0.2787	0.2922
10^{-1}	0.5776	0.5845	0.5968	0.6103
1	1.286	1.293	1.305	1.319
10	2.350	2.357	2.370	2.383
10^2	3.492	3.499	3.511	3.525
10^3	4.642	4.649	4.662	4.675
10^4	5.794	5.801	5.813	5.826
10^5	6.945	6.952	6.964	6.978

Influence of r_w on Well Response. The results presented in Figs. 2 through 4 assume that $r_{wD}=10^{-4}$. We have chosen this value of r_{wD} principally because we believe that it represents the value that will be encountered in practice, particularly for long horizontal wells. In field applications, the wellbore diameter will be dictated by the drilling constraints but will not vary by more than a factor of two. To evaluate the influence of changes in wellbore radius, we computed pressure responses for r_{wD} values in the range of $5\times10^{-5}\leq r_{wD}\leq 5\times10^{-4}$. Typical responses for two values of L_D, $L_D=10$ and $L_D=1$, are shown in Tables 1 and 2, respectively. Considering the results in Table 1, we note that the differences are significant at early times. (In actual practice, the difference at early times may not be as large as that shown here. We would expect the differences to be smaller for a well of finite radius.) Long-term productivity, however, is unaffected. For example, at $t_D=10^4$ the change in dimensionless pressure is only 1.3%. For small values of L_D (Table 2), r_{wD} has a greater influence on long-term productivity. For the ranges considered in Table 2, dimensionless pressures vary by approximately 10%. The results presented cover a wider range of r_{wD} values and for a given value of L_D, one will not encounter such large variations in r_{wD}. The tabulations are presented to emphasize the importance of r_{wD}, particularly on the early-time pressure response.

Influence of z_{wD} on Well Response. Tables 3 and 4 provide information on the influence of the well location, z_{wD}, for two values of L_D ($L_D=25$ and $L_D=0.25$, respectively). In each table, four horizontal-well locations are considered: $z_{wD}=0.5$, 0.25, 0.125, and 0.0625. The tabulations show that for these L_D values, the pressure responses are insensitive to well location. These results suggest that the pseudoskin factor will be essentially independent of z_{wD}. This result may be explained if we note that for large values of L_D, the horizontal-well solutions are essentially identical to the vertical-fracture solution; i.e., flow in the vertical direction is negligibly small. Consequently the location of the wellbore within the productive interval is not significant. If we compare pressure responses for a smaller value of L_D (Table 4), then the pressure responses are independent of z_{wD} at early times. This result can be attributed to the fact that for small values of L_D, the initial radial-flow period exists for a very long time. At later times, differences become significant and at $t_D=10^5$ the pressure drops differ by about 8% (for $L_D=25$, the difference at $t_D=10^5$ is $<0.5\%$). Although differences in p_{wD} values at late times are much larger than the differences in Table 3, the influence of z_{wD} on the pseudoskin factor may not be significant because the magnitudes of the dimensionless wellbore pressures during the pseudoradial-flow period for small L_D values are much larger than the magnitudes of the dimensionless wellbore pressures for the vertical-fracture solution at corresponding times. Thus, we expect the percent change in pseudoskin factor to be of the same order of magnitude as the dimensionless wellbore pressure. It is interesting that in this case the pressure response is relatively independent of z_{wD} for $0.125\leq z_{wD}\leq 0.875$. The effect of r_{wD} on the pressure response for various well locations is similar to that discussed in the previous section.

Pseudoskin Factor. As noted earlier, pseudoskin factors of horizontal wells are calculated by comparing their late-time pressure responses with that of a fully penetrating vertical fracture. Fig. 5 shows the variation in S as a function of L_D ($z_{wD}=0.5$). The variable of interest is r_{wD}, for which we have considered values

TABLE 4—THE INFLUENCE OF WELL LOCATION ON PRESSURE RESPONSE OF AN INFINITE-CONDUCTIVITY HORIZONTAL WELL ($L_D=0.25$, $r_{wD}=10^{-4}$)

	Dimensionless Wellbore Pressure, p_{wD}			
Dimensionless	Dimensionless Well Location, z_{wD}			
Time, t_D	0.5	0.25	0.125	0.0625
2×10^{-6}	6.109	6.109	6.109	6.109
1×10^{-5}	7.717	7.717	7.717	7.717
10^{-4}	10.02	10.02	10.02	10.02
10^{-3}	12.32	12.32	12.32	12.32
10^{-2}	14.61	14.61	14.61	14.61
10^{-1}	16.58	16.58	16.60	16.92
1	17.99	18.10	18.59	19.42
10	19.07	19.36	19.99	20.87
10^2	20.21	20.50	21.14	22.02
10^3	21.37	21.65	22.29	23.17
10^4	22.52	22.80	23.44	24.32
10^5	23.67	23.95	24.59	25.47

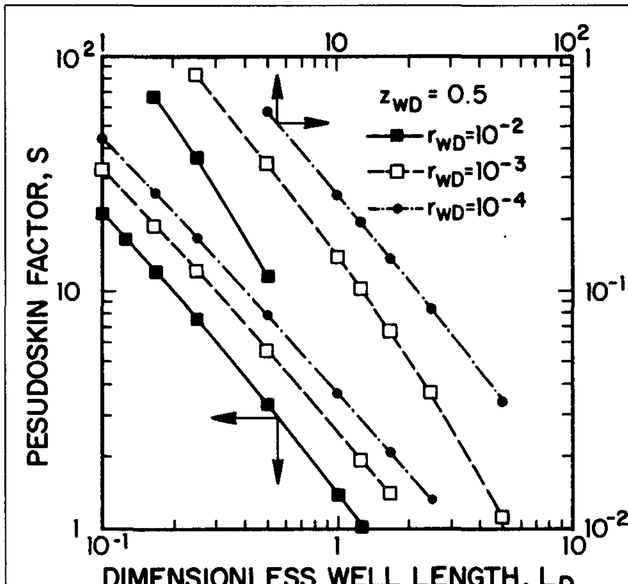

Fig. 5—Influence of dimensionless well radius on pseudoskin factor.

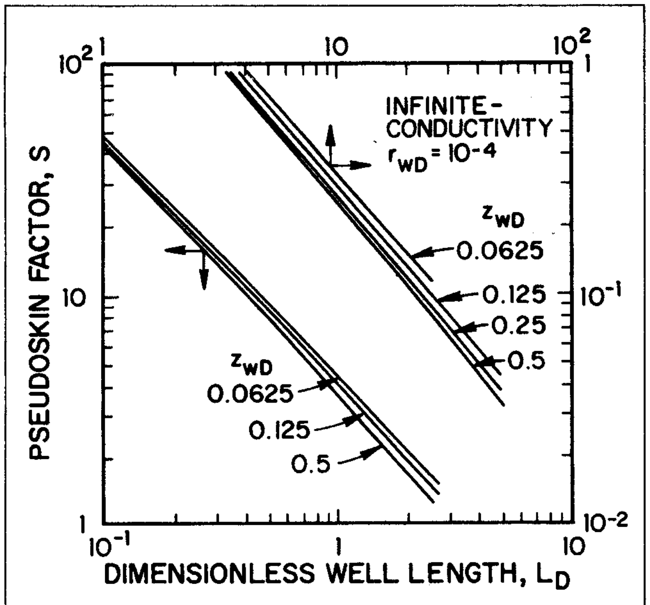

Fig. 6—Influence of well location on pseudoskin factor.

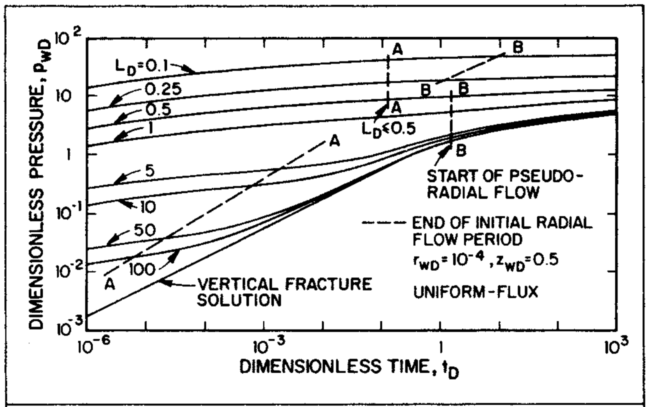

Fig. 7—Pressure response for horizontal well—uniform flux.

as large as 10^{-2} because we wish to provide solutions for short drainholes. Note, however, that if the drainhole is short, the upper limit for L_D will be much smaller than the upper limit considered in this study. As expected from the results given earlier, for a fixed value of r_{wD}, the pseudoskin factor increases as L_D decreases, and for a fixed value of L_D, pseudoskin factors decrease as r_{wD} increases. An increase in r_{wD} should be viewed as an increase in the wellbore diameter, all other variables being fixed. Note that if a large enough r_{wD} is chosen, the pseudoskin factors would be negative for some values of L_D. Under the assumptions of this work, this result is obtained because well responses are computed at different points for the horizontal-well and vertical-fracture solutions. (Actually, one may view the vertical-fracture solutions in the literature to be computed at $r_{wD}=0$.) We believe that these results are unrealistic and have not considered them in this work.

Fig. 6 depicts the variation in the pseudoskin factor, S, for various values of z_{wD} ($r_{wD}=10^{-4}$). Similar results are obtained for other values of r_{wD}. For large values of L_D, pseudoskin factors are negligibly small and the well location does not affect the magnitude of the pseudoskin factor; however, the variation in pseudoskin factor is 10%. For smaller values of L_D, we note that the pseudoskin factors are large; for example, for a well located at $z_{wD}=0.5$, $S \approx 40.5$ for $L_D=0.1$. The additional pressure drops caused by the proximity of the upper or lower boundary does not appear to be significant. Large values of pseudoskin factors for small values of L_D may be explained as follows. For simplicity, if we assume an isotropic reservoir, then values of $L_D \leq 0.5$ represent well lengths that are less than the reservoir height. Such wells can be viewed as limited-entry vertical wells, and in such cases, large pseudoskin factors are reasonable.

Although computation of the pseudoskin factor by Eq. 20 is simple, we have correlated pseudoskin factors for both infinite-conductivity and uniform-flux cases. These correlations are given in the original version of this paper.[17]

Influence of L on Well Response. As is evident from the results presented thus far, the well length has a dominant effect on well productivity. If we wish to examine the influence of length on productivity, then some care should be taken because a change in well length influences both r_{wD} and L_D, all other variables being constant. For example, if we consider a system with $L_D=10$ and $r_{wD}=5\times10^{-5}$ (Col. 2, Table 1) and wish to examine the effect of reducing L by a factor of 10, then we must consider the solution for $L_D=1$, $r_{wD}=5\times10^{-4}$ (Col. 4, Table 2). If we compare responses for these two cases, we find that the pseudoskin factor changes from 0.2881 to 2.857; a change of about 900%.

Uniform-Flux Solutions

From the responses of vertically fractured wells, we expect the characteristics of the response curves for the uniform-flux solutions to be similar to those of the response curves for the infinite-conductivity case. Fig. 7 presents solutions for the uniform-flux case. The well is assumed to be at the center of the reservoir ($z_{wD}=0.5$). If we compare the magnitude of the pressure drops with the corresponding cases in Fig. 2, we find that they are generally greater than or equal to the corresponding infinite-conductivity solution. During the initial radial-flow period, the responses would be identical because the conductivity has no influence on the well response. The dashed lines (lines AA) denote the end of the initial radial-flow period. The duration of this period was determined by computing the slope of the pressure trace. For $L_D \geq 1$, these times are essentially identical to the times for the infinite-conductivity case. For intermediate values of L_D, $0.5<L_D\leq1$, this flow period exists for a longer time.

The lines (lines BB) denote the beginning of the pseudoradial-flow period. As in the infinite-conductivity case, the beginning of this flow period is independent of L_D if $L_D \geq 0.5$. In this case, the

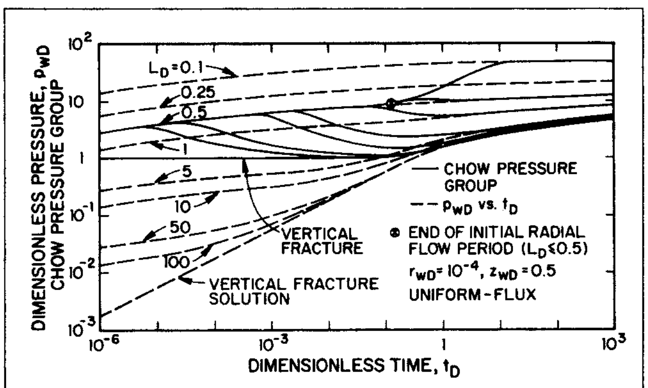

Fig. 8—Composite type curve, pressure response, and derivative response—uniform flux.

TABLE 5—DIMENSIONLESS PRESSURE AT $t_D=10^{-5}$, AT $x_D=0$ AND $|x_D|=1$ (Uniform-Flux Solution; $z_{wD}=0.5$, $r_{wD}=10^{-4}$)

Dimensionless Well Length, L_D	Dimensionless Pressure, p_{wD}: Well Center, $x_D=0$	Well Tip, $x_D=1$
0.1	53.263	37.00
1.0	10.85	8.988
10.0	7.415	6.662
50.0	7.196	6.499
Vertical Fracture	7.161	6.468

pseudoradial-flow period begins earlier ($t_D \approx 1.6$ compared with 4.13 for the infinite-conductivity case). For $L_D < 0.5$, the upper and lower boundaries also determine the start of the pseudoradial-flow period, and, thus, these times depend on L_D.

Fig. 8 is a composite of pressure and derivative responses in terms of the Chow pressure group. The characteristics of these derivative curves are similar to those considered in Fig. 4. The circle circumscribing the letter "x" on this plot denotes the end of the initial radial flow for $L_D \leq 0.5$. At late times, the normalized-pressure curves merge with the pressure-response curves (Eq. B-25 in Ref. 9). The normalized-pressure plots should enable us to identify the appropriate radial-flow periods and to choose the appropriate L_D curve to match pressure data if L_D is small. The type-curve-matching procedure is identical to that discussed earlier.

Because the characteristics of the well responses for the uniform-flux case are similar to those for the infinite-conductivity case, it is not worthwhile to document the influence of z_{wD}, r_{wD}, etc. in detail.

As mentioned earlier, computation of responses at the well tip do not appear to be worthwhile for the uniform-flux case. Table 5 compares well responses at the well center, $x_D=0$, and at the well tip, at $t_D=10^5$. The results demonstrate that the dimensionless pressures at the well tips are less than those at the well center (see also Eq. 8). Thus, for a single drainhole or a single horizontal well where produced fluids are pumped out from the well tip, the uniform-flux assumption is an unrealistic boundary condition because the point in a wellbore from where fluid is pumped to the surface must be the point of lowest pressure within the well. The infinite-conductivity assumption is the only viable boundary condition for single drainholes or horizontal wells. Because the pressure at the well center is the lowest, the uniform-flux boundary condition can be used as an alternative to the infinite-conductivity idealization for two equal-length drainholes that are drilled in diametrically opposite directions from a single vertical well.

The Effective Wellbore Radius. To compare vertical-well vs. horizontal-well productivities, it is useful to define an effective wellbore radius, which is the wellbore radius of an unstimulated vertical well, to yield the same productivity as that of a horizontal well. The concept of effective wellbore radius was used by Prats[18] and Prats *et al.*[19] to show that the behavior of infinite-conductivity vertical fractures can be represented by the behavior of an unstimulated vertical well with wellbore radius equal to one-quarter of the fracture length. If we consider 3D anisotropy ($k_x \neq k_y \neq k_z \neq k_x$), then combining Eqs. 11 and 20, we can write:

$$p_{wD}=0.5(\ln \tilde{t}_D+0.80907)$$
$$-0.5\ln\frac{(L/2)^2\{4\sin\pi[z_{wD}+(r_{wzD}/2)]\sin\pi(r_{wzD}/2)\}^{\sqrt{k_x/k}/L_D}}{r_w'^2\exp[2(1+\sigma-\varphi)]}, \quad (29)$$

where $\tilde{t}_D=t_D(L/2r_w')^2$. . . . . (30)

The late-time response of an unstimulated vertical well is given by

$$p_{wD}=0.5(\ln t_D+0.80907). \quad (31)$$

Comparing Eq. 29 with Eq. 31, we can decide that a horizontal well is equivalent to a vertical well with wellbore radius given by

$$r_w'=\frac{L/2}{\exp(1+\sigma-\varphi)}\left\{4\sin\left[\frac{\pi}{h}\left(z_w+\frac{r_w}{2}\sqrt{\frac{k_z}{k_y}}\right)\right]\times\sin\left(\frac{\pi}{2h}r_w\sqrt{\frac{k_z}{k_y}}\right)\right\}^{(h/L)\sqrt{k_x/k_z}} \quad (32)$$

Note from Eqs. 20 and 24 that for $L_D \geq 1.15/(\sqrt{k/k_x}-|x_D|)$, we have $\varphi \approx 0$, and Eq. 32 can be written as

$$r_w'=\frac{L/2}{\exp(1+\sigma)}\left\{4\sin\left[\frac{\pi}{h}\left(z_w+\frac{r_w}{2}\sqrt{\frac{k_z}{k_y}}\right)\right]\times\sin\left(\frac{\pi}{2h}r_w\sqrt{\frac{k_z}{k_y}}\right)\right\}^{(h/L)\sqrt{k_x/k_z}} \quad (33)$$

Note also from Eq. 20 that as $L_D \to \infty$, $F \to 0$; horizontal-well pressure response approaches the vertically fractured well response, and Eq. 32 yields

$$r_w'=(L/2)/\exp(1+\sigma), \quad (34)$$

where $1+\sigma=1$ and ln 2 for uniform-flux and infinite-conductivity conditions, respectively. Eq. 34 defines the effective wellbore radius for a vertically fractured well (see Refs. 8, 18, and 19). An expression similar to that given by Eq. 33 has been obtained in Refs. 20 and 21 from an approximate solution for long horizontal wells.

Conclusions

1. Horizontal-well pressure responses are a function of the dimensionless well length, L_D, and the dimensionless well radius, r_{wD}.
2. For a single horizontal well or a drainhole, the infinite-conductivity idealization is the only viable boundary condition. For two drainholes drilled in diametrically opposite directions from a single vertical well, either the infinite-conductivity or the uniform-flux idealization is appropriate.
3. If $L_D > 25$, then the long-time ($t_D > 10$) pressure response of a horizontal well is essentially identical to that of a fully penetrating vertical fracture. If $L_D \geq 50$, the pressure response of a horizontal well is indistinguishable from that of a vertically fractured well for $t_D > 10^{-2}$. Our computations suggest that pseudoskin factors may be assumed to be negligible if $L_D \geq 4$.
4. The pressure response of horizontal wells and pseudoskin factors are insensitive to the well location in the vertical plane of the reservoir.

Nomenclature

a = arbitrary constant
b = arbitrary constant
B = FVF, ft³/ft³ [m³/m³]
c_t = total compressibility, psi^{-1} [kPa^{-1}]
F = defined in Eq. 13
F' = defined in Eq. A-3
h = reservoir thickness, ft [m]
k = horizontal permeability, md
k_x = permeability in x direction, md
k_y = permeability in y direction, md
k_z = permeability in z direction, md
Ki_1 = integral of $K_0(x)$
$K_0(x)$ = modified Bessel function of second kind, order zero (Eq. 13)
L = horizontal well length, ft [m]
L_D = dimensionless well length (Eq. 7)
n = positive integer
p = pressure, psia [kPa]
p_D = dimensionless pressure (Eq. 2)
p_{fD} = dimensionless fracture pressure
p_i = initial reservoir pressure, psia [kPa]
p_{wD} = dimensionless wellbore pressure
q = flow rate, STB/D [stock-tank m³/d]
r = radial distance, ft [m]
r_D = dimensionless radial distance
$\tilde{r}_D$ = dimensionless distance defined by Eq. 14
r_w = wellbore radius, ft [m]
r_w' = effective wellbore radius, ft [m]
r_{wD} = dimensionless wellbore radius (Eq. 18)
r_{wzD} = dimensionless wellbore radius (Eq. 19)
s = Laplace variable
S = pseudoskin factor
t = time, hours or days
t_D = dimensionless time (Eq. 3)
$\tilde{t}_D$ = dimensionless time based on r_w'
x = distance in x direction (Fig. 1), ft [m]
x_D = dimensionless distance in x direction, Eq. 4
y = distance in y direction (Fig. 1), ft [m]
y_D = dimensionless distance in y direction, Eq. 5
y_w = location of well center in y direction
z = distance in z direction (Fig. 1), ft [m]

z_D = dimensionless distance in z direction, Eq. 6
z_w = well location (Fig. 1), ft [m]
z_{wD} = dimensionless well location
β = see Eq. 8
δ_D = dimensionless distance defined in Eq. 9
η_j = diffusivity constant, $j=x$, y, or z
μ = fluid viscosity, cp [mPa·s]
$\sigma(x_D,y_D)$ = defined in Eq. 10
ϕ = formation porosity, fraction
φ = defined in Eq. 21

Superscripts

$-$ = average
$'$ = derivative

Acknowledgments

We thank the management of Phillips Petroleum Co. for permission to publish this work. The assistance of R. Heiser and C.D. Javine of Phillips in the statistical analysis and data preparation, respectively, is gratefully acknowledged. Portions of this work will appear in the PhD dissertation of E. Ozkan, who acknowledges the financial support of the Ministry of Natl. Education, Ankara, Turkey.

References

1. Vicanek, J.: "Oil Production with Horizontal Wells," *Erdoel-Erdgas* (March 1985) **101**, No. 3, 85-90.
2. Giger, F.M., Reiss, L.H., and Jourdan, A.P.: "The Reservoir Engineering Aspects of Horizontal Drilling," paper SPE 13024 presented at the 1984 SPE Annual Technical Conference and Exhibition, Dallas, Sept. 16-19.
3. Daviau, F. *et al.*: "Pressure Analysis for Horizontal Wells," *SPEFE* (Dec. 1988) 716-24; *Trans.*, AIME, **285**.
4. Goode, P.A. and Thambynayagam, R.K.M.: "Pressure Drawdown and Buildup Analysis of Horizontal Wells in Anisotropic Media," *SPEFE* (Dec. 1987) 683-97; *Trans.*, AIME, **283**.
5. Clonts, M.D. and Ramey, H.J. Jr.: "Pressure-Transient Analysis for Wells With Horizontal Drainholes," paper SPE 15116 presented at the 1986 SPE California Regional Meeting, Oakland, April 2-4.
6. Chow, V.T.: "On the Determination of Transmissibility and Storage Coefficients from Pumping Test Data," *Trans.*, AGU (1952) **33**, 397-404.
7. Gringarten, A.C. and Ramey, H.J. Jr.: "The Use of Source and Green's Functions in the Solution of Unsteady-Flow Problems in Reservoirs," *SPEJ* (Oct. 1973) 285-96; *Trans.*, AIME, **254**,
8. Gringarten, A.C., Ramey, H.J. Jr., and Raghavan, R.: "Unsteady-State Pressure Distributions Created by a Well With a Single Infinite-Conductivity Vertical Fracture," *SPEJ* (Aug. 1974) 347-60; *Trans.*, AIME, **257**.
9. Ozkan, E., Raghavan, R., and Joshi, S.D.: "Supplement to SPE 16378, Horizontal-Well Pressure Analysis," paper SPE 20271 available from SPE Book Order Dept., Richardson, TX.
10. Ozkan, E.: "Performance of Horizontal Wells," PhD dissertation, U. of Tulsa, Tulsa, OK (1988).
11. Rosa, A.J. and Carvalho, R. de S.: "A Mathematical Model for Pressure Evaluation in an Infinite-Conductivity Horizontal Well," *SPEFE* (Dec. 1989) 559-66.
12. Reiss, L.H.: "Production From Horizontal Wells After Five Years," *JPT* (Nov. 1987) 1411-16; *Trans.*, AIME, **283**.
13. Joshi, S.D.: "Augmentation of Well Productivity With Slant and Horizontal Wells," *JPT* (June 1988) 729-39; *Trans.*, AIME, **235**.
14. Raghavan, R.: "Pressure Derivatives: Their Use in Well Test Analysis," paper SPE 17141 available from SPE, Richardson, TX.
15. Giger, F.M.: "Horizontal Wells Production Techniques in Heterogeneous Reservoirs," paper SPE 13710 presented at the 1985 SPE Middle East Oil Show, Bahrain, March 11-14.
16. Rodriguez, F., Horne, R.N., and Cinco-Ley, H.: "Partially Penetrating Vertical Fractures: Pressure-Transient Analysis of an Infinite-Conductivity Fracture," paper SPE 12743 presented at the 1984 SPE California Regional Meeting, Long Beach, April 11-13.
17. Ozkan, E., Raghavan, R., and Joshi, S.D.: "Horizontal Well Pressure Analysis," paper SPE 16378 presented at the 1987 SPE California Regional Meeting, Ventura, April 8-10.
18. Prats, M.: "Effect of Vertical Fractures on Reservoir Behavior—Incompressible Fluid Case," *SPEJ* (June 1961) 105-18; *Trans.*, AIME, **222**.
19. Prats, M., Hazebroek, P., and Strickler, W.R.: "Effect of Vertical Fractures on Reservoir Behavior—Compressible Fluid Case," *SPEJ* (June 1962) 87-94; *Trans.*, AIME, **225**.
20. van der Vlis, A.C., Duns, H., and Luque, R.F.: "Increasing Well Productivity in Tight Chalk Reservoirs," *Proc.*, 10th World Pet. Cong., Bucharest (1979) **3**, 71-78.
21. Giger, F., Combe, J., and Reiss, L.H.: "L'Intérêt du forage horizontal pour l'exploitation de gisements d'hydrocarbures," *Revue Inst. Français Du Pétrole* (May-June 1983) **38**, No. 3, 329-50.
22. Kuchuk, F.J. and Kirwan, P.A.: "New Skin and Wellbore Storage Type Curves for Partially Penetrated Wells," *SPEFE* (Dec. 1987) 546-54; *Trans.*, AIME, **290**.
23. Gradshteyn, I.S. and Ryzhik, I.M.: *Table of Integrals, Series, and Products*, Academic Press Inc., Orlando, FL (1980) 40.
24. Jones, P.: "Reservoir Limit, Well and Formation Evaluation," paper 824-G presented at the 1957 SPE Permian Basin Oil Recovery Symposium, Midland, April 18-19.
25. Jones, P.: "Reservoir Limit Test on Gas Wells," *JPT* (June 1962) 613-19.
26. Carter, R.D.: "Pressure Behavior of a Limited Circular Composite Reservoir," *SPEJ* (Dec. 1966) 328-33; *Trans.*, AIME, **237**.
27. Tiab, D. and Kumar, A.: "Application of the p_D' Function to Interference Analysis," *JPT* (Aug. 1980) 1465-70.
28. Bourdet, D. *et al.*: "A New Set of Type Curves Simplifies Well-Test Analysis," *World Oil* (May 1983) 95-106.
29. Onur, M. and Reynolds, A.C.: "A New Approach for Constructing Derivative Type Curves for Well Test Analysis," *SPEFE* (March 1988) 197-206; *Trans.*, AIME, **285**.
30. Duong, A.N.: "A New Set of Type Curves for Well Test Interpretation With the Pressure/Pressure-Derivative Ratio," *SPEFE* (June 1989) 264-72.

SPEFE

Original SPE manuscript received for review April 8, 1987. Paper accepted for publication April 3, 1989. Revised manuscript received Nov. 30, 1988. Paper (SPE 16378) first presented at the 1987 SPE California Regional Meeting held in Ventura, April 8-10.

SPE 20271, "Supplement to SPE 16378, Horizontal-Well Pressure Analysis," available from SPE Book Order Dept.